



Fine-grained vehicle damage classification based on applicability network

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HIGHLIGHTS

- A universally applicable network for damage and defect classification, FGDA Net, is proposed.
- Independent feature extraction (IR Layer) and regional learning modules (RL Block) are introduced.
- An adaptive global feature attention algorithm (GFS Module) is proposed.
- The generalized feature constraint loss (GFC Loss) is introduced.
- The network achieves excellent performance in vehicle damage classification and steel surface defect classification.

ARTICLE INFO

Keywords:

Vehicle intelligent damage assessment
Damage defect classification
Fine-grained learning
Backbone
Regularization correction

ABSTRACT

Vehicle damage classification is a critical component of intelligent damage assessment systems, presenting challenges due to complex damage morphology, high inter-class similarity, and significant intra-class variance. To address these issues, this paper proposes a Fine-grained Defect Applicable Network (FGDA Net) that enhances classification accuracy through a multi-level fine-grained feature perception mechanism. Specifically, FGDA Net alternately introduces the In-depth & Refined Layer (IR Layer) and the Regional Learning Block (RL Block) to perform independent down-sampling and fine-grained feature attention, thereby reducing information loss and strengthening the perception of critical regions. Furthermore, a Global Feature Search Module (GFS Module) is designed to adaptively search and aggregate global self-similar information within shallow networks, improving inter-class discrimination. Additionally, a Generalized Feature Correction Loss (GFC Loss) is proposed to regularize the backbone network through reverse feature fusion and label smoothing, enhancing feature generalization. On an 8-class vehicle damage dataset, FGDA Net achieves a 4.32% improvement in Top-1 classification accuracy over the baseline model, outperforming several advanced classification networks. The method also demonstrates strong generalization on a public steel plate surface defect dataset. Experimental results indicate that FGDA Net effectively addresses the challenges of damage defect classification and is suitable for real-world intelligent damage assessment and industrial quality inspection scenarios.

1. Introduction

Vehicles have become one of the indispensable means of transportation [1]. However, with the widespread use of automobiles, traffic accidents inevitably occur frequently, leading to a growing demand for intelligent damage assessment [2]. Specifically, conventional manual on-site damage assessment suffers from drawbacks such as high costs, lack of standardization, and low efficiency, making it difficult to meet the core requirements for processing safety [3,4] and efficiency [5]. In contrast, intelligent damage assessment, which integrates deep learning with vehicle damage evaluation, significantly enhances both the

safety and efficiency of processing accident-damaged vehicles. The typical pipeline of intelligent damage assessment is as follows: images of vehicle damage are uploaded to the assessment system, which then automatically categorizes the types of damage, detects the corresponding components, and ultimately provides a compensation plan. It can be seen that achieving rapid and accurate classification of vehicle damage is the key technical foundation for the efficient operation of intelligent damage assessment systems.

Current deep learning-based classification networks primarily adopt three main frameworks: Convolutional Neural Networks (CNNs) [6,7],

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<https://doi.org/10.1016/j.asoc.2026.115076>

Received 12 August 2024; Received in revised form 23 February 2026; Accepted 18 March 2026

Available online 19 March 2026

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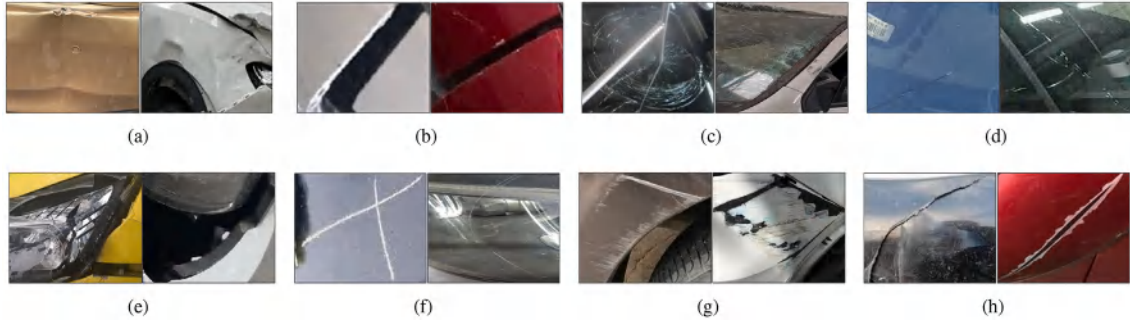


Fig. 1. Eight common types of vehicle damage: two distinct representative data images are presented for each class, (a) deformation, (b) dislocation, (c) glass breakage, (d) glass crack, (e) missing, (f) scratch, (g) scratches, and (h) tearing.

the Transformer [8], and the Mamba framework [9]. Common CNN-based networks [10–13] demonstrate strong local feature representation and translation invariance when applied to natural scene objects, providing significant advantages in extracting low-level features and visual structures [14,15]. However, unlike objects in natural scenes, the inter-class differences among damage defect targets are smaller. The features learned by general-purpose CNN frameworks are influenced by their local representations, making it difficult to meet the requirements for fine-grained vehicle damage classification. While the Transformer framework [16] excels in capturing long-range and interdependent features in images compared to traditional CNN algorithms, it requires large-scale training data. Industrial defect datasets, such as those for vehicle damage, often struggle to meet these data requirements due to the challenges in acquiring defect images. Therefore, the Transformer framework also struggles to achieve good results in industrial applications such as vehicle damage assessment. For the Mamba framework, its design based on the State Space Model (SSM) offers linear computational complexity and strong long-sequence modeling capabilities, enabling it to effectively model global contextual information while overcoming the Transformer framework's reliance on massive amounts of data. However, in fine-grained damage classification tasks, the inductive bias of SSM for local features is weak, making it difficult to effectively model subtle local differences.

The challenges faced by the aforementioned frameworks in handling vehicle damage classification stem from the inherent visual complexity of the task itself. As shown in Fig. 1, on one hand, different classes of damage may exhibit high visual similarity. For instance, both glass cracks and scratches appear as elongated, linear defects; dislocation and tearing both manifest as dark, strip-like voids; and missing components such as side mirrors or windows are often accompanied by extensive glass breakage, which can easily be confused with glass breakage itself. On the other hand, the same damage class may show significant variation across different images. For example, deformation can range from small, localized areas to extensive surface buckling; the distribution of fragments and the intensity of reflections in glass breakage vary widely; and scratches differ in area, depth, orientation, and distribution patterns depending on the material and applied force, displaying highly diverse characteristics. In summary, as depicted in Fig. 2, vehicle damage classification presents challenges in distinguishing similar morphologies between classes and identifying complex morphologies within the same class.

In existing research on damage defect classification [17–19], most approaches introduce attention mechanisms into the network feature learning process [20,21], or employ holistic scanning convolution modules for multi-dimensional global modeling of the image background [22,23], to enhance model understanding and performance in complex scenes. However, these methods lack awareness of fine-grained target features, making it difficult to achieve intra-class clustering and inter-class separation in the feature space, which ultimately leads to misdetection.

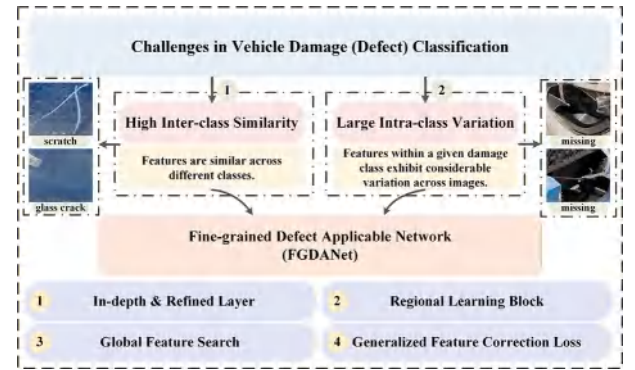


Fig. 2. The problems existing in the classification of vehicle damage and the solutions provided by the model proposed in this paper.

Therefore, as illustrated in Fig. 2, we propose a novel framework that leverages multiple fine-grained feature perception modules to learn critical features of damaged regions. First, we decompose feature down-sampling and feature learning into two independently implemented modules to reduce information loss and noise interference, and employ residual connection-based feature refinement learning to achieve fine-grained feature extraction. Simultaneously, within the shallow networks, we explore global information by aggregating relevant features from the input feature maps to highlight strongly discriminative key regions. Finally, through loss function computation, we regularize the parameters of the backbone network to enhance its generalization capability. Our main contributions are as follows:

- 1) A Fine-grained Defect Applicable Network (FGDANet) is proposed. This network significantly improves the accuracy of vehicle damage classification by introducing multiple synergistically operating fine-grained perception modules. FGDANet also exhibits strong generalization capability on steel plate surface defect data, validating its applicability to industrial defect classification tasks.
- 2) The In-depth & Refined Layer (IR Layer) and Regional Learning Block (RL Block) are designed and alternately introduced. The IR Layer is responsible for independent feature mapping and down-sampling, while the RL Block, through its Region Detail Module and Residual Learning Module, achieves fine-grained feature extraction and enhancement of critical regions, strengthening the model's local perception capability for damaged areas.
- 3) A Global Feature Search Module (GFS Module) is proposed. Embedded within the shallow networks, this module adaptively aggregates global self-similar information through a semantic fingerprint-guided similar feature search mechanism, assisting

Table 1
Summary of related work.

Research direction	Specific approach	Core idea/contribution	Limitation
Damage defect research based on deep learning	Model structure adjustment	Introducing attention mechanisms and deformable convolutions, focusing on macro-level improvements to the model architecture.	Does not emphasize the intrinsic learning capability of basic classification networks for fine-grained defect features.
	Multi-stage detection	Employing multi-stage processing pipelines.	Complex pipeline with high computational cost; typically targets detection/segmentation tasks rather than directly optimizing fine-grained discriminative power for classification.
	Artificial sample generation	Generating artificial defect samples and annotations to address data scarcity and imbalance issues.	Generation quality and diversity rely on large amounts of data or complex training.
Fine-grained classification research based on deep learning	Backbone network optimization	Introducing attention mechanisms to assist feature integration, or combining lightweight convolutions with scanning strategies to simulate complex defect morphologies.	Lacks finer-grained attention to critical regions and full utilization of detailed features.
	Loss regularization correction	Directly guiding the network to focus on salient regions.	Lacks systematic integration and utilization of features already extracted by the network.

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the model in attending to discriminative fine-grained regions at an early feature stage, thereby enhancing inter-class discrimination.

- 4) A Generalized Feature Correction Loss (GFC Loss) is proposed. Building upon the cross-entropy loss, it introduces reverse-fused multi-layer features and smoothed labels to compute a weakly supervised loss, thereby achieving generalization regularization correction of the backbone network parameters. This enhances the model's ability to learn features from negative samples and improves the network's robustness.

The organizational structure of this paper is as follows: [Section 2](#) provides an overview of the related work on damage defect research and fine-grained classification. [Section 3](#) details the implementation of the four modules proposed in this paper. [Section 4](#) presents the experimental results. Finally, [Section 5](#) summarizes this paper.

2. Related work

This section reviews and organizes existing related work from two perspectives: damage defect research and fine-grained image classification, with a summary provided in [Table 1](#).

2.1. Research on damage defects based on deep learning

Damage defect targets commonly face challenges of high inter-class similarity and significant intra-class variance. Current research addressing these issues can be broadly categorized into three main approaches: model adjustments [24–27], multi-stage detection [28–30], and artificial sample generation [31–33]. For instance, the LSwIn Transformer model [34] combines the global dependency construction capability of the Swin Transformer [35] with the feature extraction capability of CNNs, enhancing the model's overall perception of defect features. Works [36,37] introduce attention mechanisms to enhance inter-channel feature interaction, effectively capturing spatiotemporal semantic relationships between images. Work [38] utilizes spatial context pyramids and deformable convolution structures to improve detection accuracy for minor damage. The aforementioned approaches represent defect detection algorithms based on model adjustments.

In the study of multi-stage detection for industrial defects, Qiu et al. [29] proposed a three-stage supervised segmentation method. Initially, the input image is preliminarily segmented using Fully Convolutional

Networks (FCN) [39]. This is followed by a secondary discrimination of the regions of interest identified in the first stage to eliminate false positives. Finally, the segmented regions are refined using guided filters. Works [40,41] first determine whether a vehicle is damaged, then locate and crop the damage based on an object detection model, and finally classify the severity of the damage to assess the vehicle's degree.

For the generation of artificial defect samples, to address the challenge of providing sufficient and balanced annotated datasets in real-world scenarios, some researchers have employed models such as variational auto-encoders [42] and generative adversarial networks [43] to generate artificial defect samples and their annotations, which are then used to train supervised methods. TF-IDG [44], on the other hand, utilizes gradient-guided generation to produce diverse, structurally aligned anomaly images with only a single anomaly sample.

Currently, research on damage defects based on deep learning has achieved substantial results in the three aforementioned research directions. However, limited emphasis has been placed by scholars on the learning capability of basic classification networks for fine-grained features of defect samples. Drawing inspiration from the research approaches of current researchers on object classification in natural scenes [45,46], this paper proposes a fine-grained classification network tailored for defects to achieve more accurate damage defect classification.

2.2. Research on fine-grained classification based on deep learning

Fine-grained image classification [47] involves recognizing sub-classes within a larger category, where the differences between sub-classes are subtle, but the variations within each subclass are significant, making clustering challenging [48]. This task, therefore, necessitates a more refined classification of image classes. Early approaches often employed localization-classification methods [49–51] for fine-grained learning. For example, Mask-CNN [52] initially segments and locates the target using FCN [39], then selects key convolutional feature descriptors based on the mask obtained from FCN, trains three subnetworks, and finally cascades the features of the three subnetworks to represent the features of the entire image.

Vehicle damage samples exhibit the characteristics of being difficult to distinguish between classes while displaying diverse morphologies within the same class, which underscores the significant importance of fine-grained classification in vehicle damage research. However, existing fine-grained classification models often consist of multiple subnetworks,

resulting in excessive parameterization and computational demands that hinder practical deployment. Our research focuses on the vehicle damage classification stage within intelligent damage assessment, aiming to achieve industrially applicable fine-grained damage defect classification through both backbone network optimization and regularization correction.

2.2.1. Optimization of backbone network

With the emergence of various high-performance backbone networks [10–13], an increasing number of fine-grained classification studies have focused on optimizing these networks. For instance, Qi et al. [53] incorporated attention modules into a basic CNN network and utilized a feature learning network that integrates diverse information, achieving promising results in fine-grained image classification. As Transformer technology has continuously advanced in the visual domain, Transformers have gradually been adopted as backbone networks [34,35,54] in fine-grained research. TransFG [55], built upon the Swin Transformer architecture [35], integrates all the original attention weights into the attention map, guiding the model to efficiently select image regions corresponding to differentiated features that distinguish classes. SwinFG [56] integrates the attention weights of local windows and employs shifted-window self-attention to pinpoint regions of significant feature differentiation, thereby enhancing fine-grained classification. In studies utilizing Mamba as the backbone network, OS-Mamba [22] performs multi-dimensional global modeling of image backgrounds, while SCEGamba [23] combines lightweight gated bottleneck convolutions with structure-aware scanning strategies to effectively model challenging defect morphologies and textures.

In current fine-grained classification algorithms based on optimized backbone networks, existing models tend to utilize attention mechanisms within the backbone network for feature integration but lack fine-grained attention to critical regions and comprehensive utilization of detailed features. Our network innovatively incorporates the Regional Learning Block (RL Block) and Global Feature Search Module (GFS Module) alongside independent downsampling operations, successfully achieving fine-grained learning of foreground detail information and global self-similarity information.

2.2.2. Loss regularization correction

In recent years, regularization correction methods have been extensively utilized across various visual tasks. Within the domain of fine-grained classification research, a variety of regularization loss functions from different perspectives have been gradually introduced and employed to facilitate fine-grained learning within networks [57–59]. For example, MC-Loss [60] identifies specific class-discriminative local regions by utilizing attention discriminative components and diversity partition components. SoftTriple loss [61] adopts an optimization metric learning strategy, bypassing the embedding stage of sampling and enhancing fine-grained feature learning by moderately increasing the size of the final fully connected layer. Wang et al. [62] address the issue of imbalance in fine-grained problems and introduce a new technique called data-dependent shrinkage, which establishes a fine-grained generalization boundary for imbalanced learning by capturing modified loss for different class handling methods. Work [38] employs sample-based contrastive learning to regularize training based on relationships between samples, thereby aligning it with the evaluation process. Work [63] proposes a task-agnostic regularization loss method that utilizes pre-extracted task-agnostic priors to constrain parameter updates, preventing the model from overfitting to existing tasks and enhancing its generalization capability to new tasks.

The aforementioned regularization correction methods predominantly focus on discriminative learning of salient and local regions corresponding to subtle visual features solely through the loss function. However, they lack integrated utilization of the features extracted by the network. To address this limitation, we propose a Generalized Feature Correction Loss (GFC Loss), which computes a weak logarithmic loss

using smoother non-target logits and smoothed labels. This approach enhances the network's attention to key regional features that can distinguish similar classes and improves the network's generalization ability.

3. Methodology

The overall framework of the Fine-grained Defect Applicable Network (FGDANet) proposed in this paper is depicted in Fig. 3. To achieve high accuracy in vehicle damage classification, addressing the current challenges in vehicle damage research (as shown in Fig. 2), FGDANet introduces four fine-grained feature perception modules. Fine-grained feature perception refers to the model's ability to selectively extract and enhance subtle, local, and discriminative visual patterns in images during the feature learning process. The four fine-grained feature perception modules are designed to enhance the model's capability to capture and distinguish key visual details in tasks characterized by small inter-class differences and large intra-class variations, through a multi-level, multi-module collaborative feature learning mechanism.

Specifically, the proposed FGDANet, by alternately introducing the In-depth & Refined Layer (IR Layer) (see Section 3.1 for details) and the Regional Learning Block (RL Block) (see Section 3.2 for details), directs the network to focus on those local regions in the input image that are crucial for class discrimination, enabling fine-grained feature extraction—that is, achieving fine-grained sampling and strong discriminative regional attention. Additionally, the Global Feature Search Module (GFS Module) (see Section 3.3 for details) is integrated into the shallow network to further leverage semantically rich global self-similarity information, thereby assisting in the learning of foreground features. Finally, with the assistance of the Generalized Feature Correction Loss (GFC Loss) (see Section 3.4 for details), the network learns feature knowledge from significant negative samples, thereby enhancing its generalization capability. This section provides a detailed introduction to the aforementioned four perceptual modules.

3.1. In-depth & refined layer

In the classical ResNet model, the feature map undergoes dimensional changes when a residual block is introduced for learning—specifically, the doubling of channel numbers and the reduction in height and width as a result of downsampling operations. Performing feature subsampling and learning simultaneously can introduce noise interference, leading to the loss of some information. Our objective is to achieve fine-grained learning of key features, thereby enhancing the model's capability to extract fine-grained features from the target region. Inspired by the Transformer framework, our model decomposes the residual block into the IR Layer and RL Block, dedicated to feature downsampling and feature information learning, respectively. Further elaboration on the RL Block is provided in Section 3.2.

The position of the IR Layer is depicted in Fig. 3. We incorporate the IR Layer into the model for feature mapping and expand the dimensions of the feature channels through downsampling. The IR Layer performs feature mapping from the original high-dimensional feature space to a low-dimensional feature space, where both source and target features exhibit the same distribution. The inclusion of the IR Layer enables the model to refine features in a more comprehensive manner, enhancing the capacity of other network modules to learn critical regional features. Additionally, the original residual block conducts downsampling with a stride of 2 during feature information learning, which may result in information loss due to uneven sampling. By independently implementing downsampling in the IR Layer, we ensure that the input and output feature dimensions of the subsequent RL Block are consistent. This mitigates the issue of information loss caused by changes in feature map dimensions during the feature learning process, allowing the model to better concentrate on sensitive feature regions, disregard background noise, and reduce the impact of sampling operations. Ultimately, this methodology facilitates fine-grained learning of highly discriminative

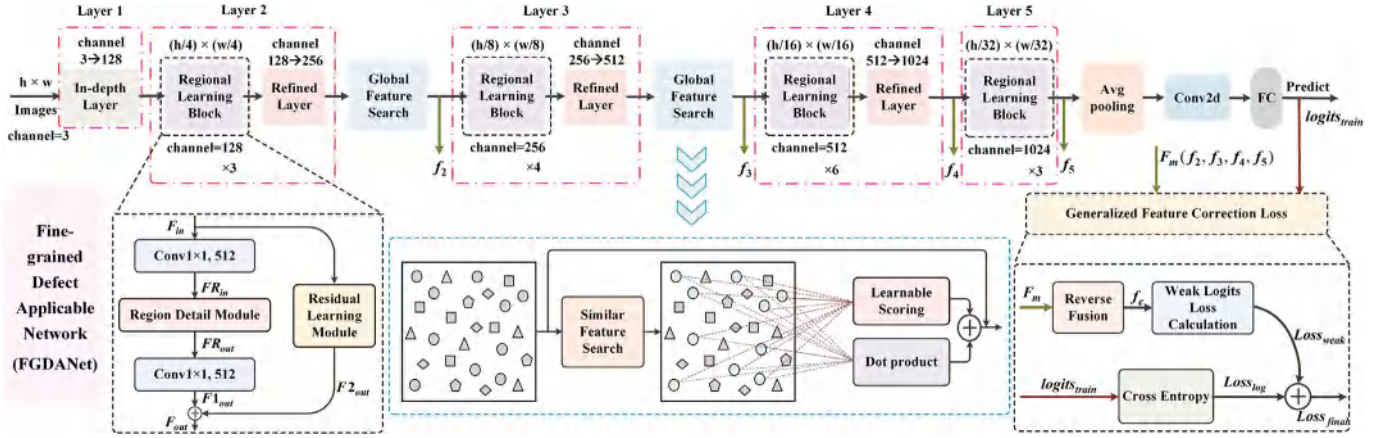


Fig. 3. The overall framework of FGDA Net: the network is divided into a five-layer feature extraction learning structure initially, consisting of alternating IR layers and RL Blocks; a GFS Module is added between shallow layers; the loss function during training is the GFC Loss, with the input features being the output features of the last four layers of the network.

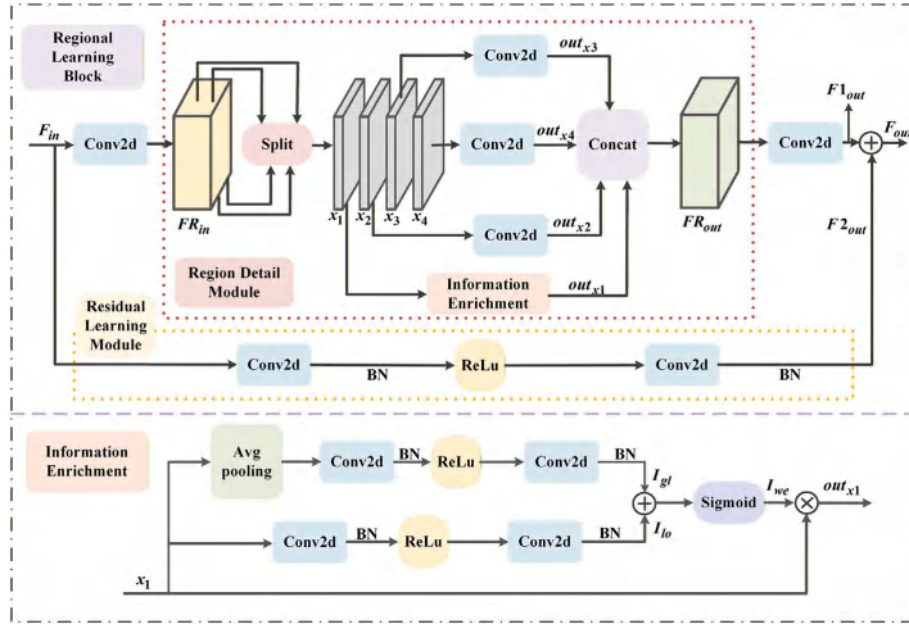


Fig. 4. The structure of the regional learning block.

regional features. The specific alterations in feature dimensions after passing through the IR Layer are detailed in Fig. 3.

3.2. Regional learning block

During the process of extracting and sensing incoming features in ResNet, residual connections are employed to learn the disparities between input and output features directly. This approach effectively mitigates the problem of gradient vanishing and, to some extent, addresses the issue of degradation in deep neural networks. Given the feasibility of residual connections and the specific requirements of damage classification, based on the feature subsampling of the IR Layer, an RL Block is proposed for the refinement learning of salient feature region information, so that the network can pay more fine-grained attention to the key feature region.

The structure of the RL Block is depicted in Fig. 4. In response to the requirement for fine-grained classification of vehicle damage, we propose the Region Detail Module and Residual Learning Module based on residual connections to achieve fine-grained learning of both global and

local feature information. In Fig. 4, the Region Detail Module is outlined by a red dashed box, and the Residual Learning Module is enclosed by a yellow dashed box. Upon the input of feature $F_{in} \in \mathbb{R}^{C \times H \times W}$ into the RL Block, it is separately passed into two forward pass pathways (mainstem forward and residual forward).

In the mainstem forward connection, feature F_{in} is initially processed through a 1×1 convolution to obtain feature $F_{R_{in}} \in \mathbb{R}^{C \times H \times W}$, which serves as the input to the Region Detail Module. In the Region Detail Module, the feature $F_{R_{in}}$ is equally divided into multiple dimensionally identical features in the channel dimension $\dim_{\text{channel}}$ through a Split operation. Here, we divide it into four identical features: x_1, x_2, x_3, x_4 ($x_i \in \mathbb{R}^{\frac{C}{4} \times H \times W}$).

$$x_1, x_2, x_3, x_4 = \text{Split}(F_{R_{in}}, \dim = \dim_{\text{channel}}) \quad (1)$$

One of the features (taking feature x_1 as an example) is passed into the Information Enrichment Module, where global and local information of foreground features is extracted to derive fine-grained sensitive region feature $\text{out}_{x1} \in \mathbb{R}^{\frac{C}{4} \times H \times W}$. After feature x_1 is passed into the Information

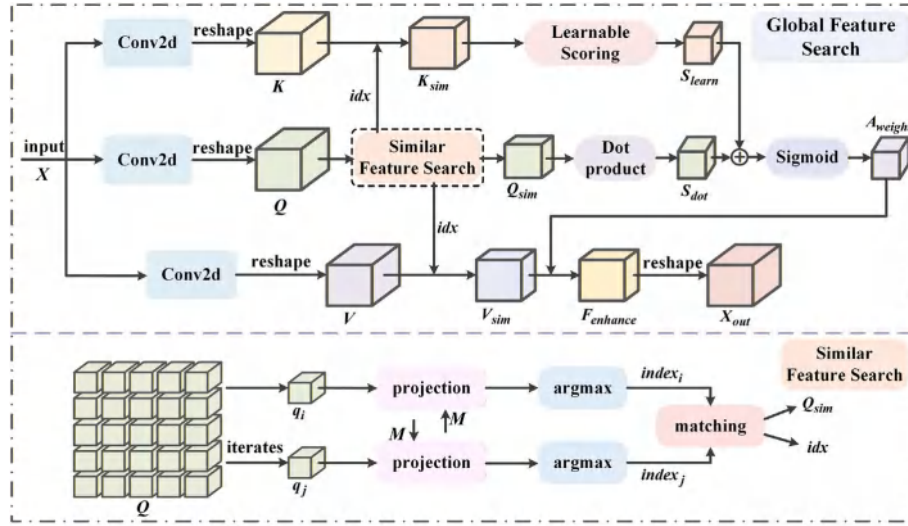


Fig. 5. The structure of global feature search.

Enrichment Module, the following operations are performed to obtain out_{x1} :

$$I_{gl} = \text{Co}(\text{Re}(\text{Co}(\text{Avg}(x_1)))) \quad (2)$$

$$I_{lo} = \text{Co}(\text{Re}(\text{Co}(x_1))) \quad (3)$$

$$out_{x1} = x_1 \cdot \delta(I_{gl} + I_{lo}) \quad (4)$$

where, Co denotes the convolution operation, Re denotes the ReLU operation, Avg denotes the global average pooling operation, δ denotes the Sigmoid operation, and I_{gl} and I_{lo} are intermediate features.

At the same time, features x_2, x_3, x_4 undergo three identical 3×3 convolution operations to obtain the corresponding output features $out_{x2}, out_{x3}, out_{x4}$ ($out_{xi} \in \mathbb{R}^{\frac{C}{4} \times H \times W}$). The four output features are concatenated to generate the feature $FR_{out} \in \mathbb{R}^{C \times H \times W}$ with an enlarged receptive field. Feature FR_{out} goes through a 1×1 convolution to get the output $F1_{out} \in \mathbb{R}^{C \times H \times W}$ of the mainstem forward connection. The aforementioned process can be represented as:

$$out_{x2} = \text{Co}(x_2), out_{x3} = \text{Co}(x_3), out_{x4} = \text{Co}(x_4) \quad (5)$$

$$FR_{out} = [out_{x1}, out_{x2}, out_{x3}, out_{x4}] \quad (6)$$

$$F1_{out} = \text{Co}(FR_{out}) \quad (7)$$

where, $[\cdot]$ denotes the concatenation operator. In the residual forward connection of RL Block, the input F_{in} is directly passed to the Residual Learning Module, where a series of convolution operations are applied to obtain local key information $F2_{out} \in \mathbb{R}^{C \times H \times W}$ corresponding to visual features. The operations of the Residual Learning Module can be represented as:

$$F2_{out} = \text{Co}(\text{Re}(\text{Co}(F_{in}))) \quad (8)$$

Finally, the RL Block output $F_{out} \in \mathbb{R}^{C \times H \times W}$ is obtained by adding the corresponding positions of $F1_{out}$ and $F2_{out}$.

The RL Block fuses the outputs of the mainstream forward path and the residual forward path through feature addition, enabling the model to simultaneously capture high-level semantic information and fine-grained local features, thereby achieving refined perception of subtle structures and discriminative features in damaged regions. Furthermore, through multi-scale feature fusion and residual connections, the model's ability to distinguish similar damage categories in the feature space

is enhanced, while features of the same damage category become more compact. This effectively alleviates the “high inter-class similarity and large intra-class variance” problem inherent in fine-grained classification.

3.3. Global feature search module

The key challenge in fine-grained visual classification lies in distinguishing categories that are difficult to differentiate using only local absolute features [64,65]. Taking vehicle damage as an example, “scratch” and “glass crack” exhibit high similarity in local morphology—both appear as linear traces, making them difficult to distinguish based solely on absolute features such as color and texture. However, the contrast between these two types of damage and their background materials differs fundamentally: scratches are embedded in a metallic painted surface, while glass cracks are embedded in a transparent, reflective glass material. This relative relationship between “damage type” and “background material” constitutes the key discriminative information for distinguishing the two. Therefore, how to effectively model such relative features is the core problem in improving fine-grained classification performance.

To address the aforementioned issue, this paper proposes a Global Feature Search Module (GFS Module), shown in Fig. 5. This module performs semantic retrieval on shallow feature maps to locate global positions that share consistent semantic attributes with the target region. Crucially, these positions include not only other points within homogeneous regions but also boundary points between the target region and the background—the receptive fields of these boundary points simultaneously cover both foreground and background, inherently encoding the contrastive information between them. By aggregating features from these boundary points and using them to enhance the original target features, the GFS Module enables the model to simultaneously model the attributes of the target itself and its contextual relationship with the background, thereby achieving an elevation in representation from absolute features to relative features.

After the shallow feature map $X \in \mathbb{R}^{C \times H \times W}$ is fed into the GFS Module, it first passes through three 3×3 convolutions to generate three features $Q \in \mathbb{R}^{C \times H \times W}$, $K \in \mathbb{R}^{C \times H \times W}$, and $V \in \mathbb{R}^{C \times H \times W}$, denoted as query, key, and value, respectively. To facilitate subsequent processing, the spatial dimensions are flattened to obtain $Q \in \mathbb{R}^{C \times N}$, $K \in \mathbb{R}^{C \times N}$ and $V \in \mathbb{R}^{C \times N}$, where $N = H \cdot W$ denotes the total number of spatial positions.

Subsequently, a Similar Feature Search is performed. For position i ($i = 1, 2, \dots, N$) in Q , its feature vector $q_i \in \mathbb{R}^C$ is first multiplied by

Algorithm 1: GFS Module.

Input:
 X : input feature ;
 M : random orthogonal rotation matrix ;
 $Anchor$: the set of top Y anchor points ;
Output:
 X_{out} : final output feature ;

```

1  $Q \leftarrow \text{reshape}(\text{Co}(X)), K \leftarrow \text{reshape}(\text{Co}(X)), V \leftarrow \text{reshape}(\text{Co}(X))$  ;
2 for  $i = 1$  to  $N$  do
3    $index_i \leftarrow \text{argmax}(M \cdot q_i)$ : After multiplying feature  $q_i$  with
   the orthogonal matrix, the index corresponding to the
   maximum value of the product is found ;
4 end
5 for  $a$  in  $Anchor$  do
6   for  $j = 1$  to  $N$  do
7     if  $index_j == index_a$  then
8        $D.append(j)$ : Point  $j$  is stored into list  $D$  ;
9     end
10  end
11   $Q_{sim} \leftarrow Q[D, :], K_{sim} \leftarrow K[D, :], V_{sim} \leftarrow V[D, :]$  ;
12   $S_{dot} \leftarrow Q_{sim} \cdot Q_{sim}^T$ : The dot product similarity is computed ;
13   $S_{learn} \leftarrow W_2(\text{Re}(W_1 \cdot K_{sim}) + b_1) + b_2$ : The learnable similarity
  is computed ;
14   $A_{weight} \leftarrow \text{Softmax}(S_{dot} + S_{learn})$ : The final attention weights
  are computed ;
15   $F_{enhance} \leftarrow A_{weight} \cdot V_{sim}$ : The enhanced feature is obtained ;
16   $X_{out}'[D, :] \leftarrow X[D, :] + F_{enhance}$ : The enhanced feature is
  mapped back to its original spatial positions in conjunction ;
17 end
18  $X_{out} \leftarrow \text{reshape}(X_{out}', (C, H, W))$ 
19 return  $X_{out}$ 

```

a random orthogonal rotation matrix $M \in \mathbb{R}^{C \times C}$ and projected into a new feature space—this operation eliminates correlations between original feature channels, enabling each channel to independently express different semantic information. Subsequently, the channel index corresponding to the maximum value in the projected vector $q_i \cdot M \in \mathbb{R}^C$ is taken as the semantic fingerprint $index_i \in \{1, 2, \dots, C\}$ for that position. Based on the preliminary localization of damaged regions by preceding modules, the GFS Module selects the top Y positions with the strongest feature responses in Q as anchor positions. It then traverses each anchor position while computing its semantic fingerprint $index_a$. Using this fingerprint as a reference, it traverses all positions across the entire feature map and records all positions with identical fingerprints. Specifically, for any position j , if its semantic fingerprint $index_j$ matches $index_a$, the feature $q_j \in \mathbb{R}^C$ at that position and its index j are stored into sets $Q_{sim} \in \mathbb{R}^{C \times L}$ and $idx \in \mathbb{R}^L$, respectively, where L denotes the number of matched positions.

Next, by indexing idx , the relevant feature sets $K_{sim} \in \mathbb{R}^{C \times L}$ and $V_{sim} \in \mathbb{R}^{C \times L}$ are obtained from K and V , respectively.

After identifying all semantically similar positions, the GFS Module needs to determine which of these positions contribute most to enhancing the anchor features. To this end, it designs two similarity computation methods. The first is dot product similarity S_{dot} , which directly computes the dot product between the anchor features and the features at each similar position:

$$S_{dot} = Q_{sim} \cdot Q_{sim}^T \quad (9)$$

where, the diagonal elements represent the self-similarity of the same position.

The second is learnable similarity S_{learn} , which introduces a learnable scoring function:

$$S_{learn} = W_2(\text{Re}(W_1 \cdot K_{sim}) + b_1) + b_2 \quad (10)$$

where, W_1 and W_2 are learnable weights, and b_1 and b_2 are bias terms.

The two similarity scores S_{dot} and S_{learn} are summed and then normalized by Softmax to obtain the final attention weights A_{weight} for each similar position. These weights are used to perform a weighted summation of V_{sim} , yielding the enhanced feature $F_{enhance} \in \mathbb{R}^{C \times L}$:

$$F_{enhance} = A_{weight} \cdot V_{sim} \quad (11)$$

Finally, in conjunction with the index idx , the enhanced feature is mapped back to its original spatial positions and reshaped to obtain the output $X_{out} \in \mathbb{R}^{C \times H \times W}$ of the GFS Module. The algorithm flow of the GFS Module is shown in Algorithm 1.

Through the GFS Module, the features at anchor positions are no longer isolated local features, but enhanced features that integrate information from all semantically similar positions across the entire feature map—they encompass both the attributes of the damaged region itself and encode the contrastive relationship between the damage and its background.

3.4. Generalized feature correction loss

During the learning phase of the backbone network, the classification network in this study achieves fine-grained extraction of salient regional features from input images through a series of downsampling operations and feature learning modules. To further leverage the feature knowledge learned from the above improvements, our model incorporates the GFC Loss during the network's loss calculation phase. The structure of GFC Loss is depicted in Fig. 6, and its algorithmic flow is detailed in Algorithm 2. Building upon cross-entropy loss, GFC Loss computes a weak logarithmic loss by backwardly fusing fine-grained features derived from the backbone network, and then realizes the regularization correction of the backbone network parameters, successfully enhancing the generalization of the network.

In the GFC Loss, the intermediate feature F_m ($f_5, f_4, f_3, f_2 \in F_m$) of the network obtain a generalized feature f_e with coarser granularity through the Reverse Fusion Module. The specific implementation process of the Reverse Fusion Module is as follows:

$$p_4 = \text{Up}(f_5) + f_4 \quad (12)$$

$$p_3 = \text{Up}(p_4) + f_3 \quad (13)$$

$$f_e = p_2 = \text{Up}(p_3) + f_2 \quad (14)$$

where, $\text{Up}(\cdot)$ denotes the upsampling operation, p_i denotes the intermediate feature value generated during the upsampling procedure. Eq. (14) enables short-path gradient propagation through skip connections and provides a feature representation that integrates multi-level semantics and spatial details for the weakly supervised branch. The output f_e of the Reverse Fusion Module further participates in the computation of the weak logits loss $\text{Loss}_{\text{weak}}$. Specifically, we incorporate weak supervision on the fused features of the network to derive a new logits. Denoting Global Average Pooling as Avg and a new linear layer as FC, where Softmax transforms the output of the FC layer into a probability distribution, the weak logits $\text{logits}_{\text{weak}}$ obtained in the Weak Logits Loss Calculation can be expressed as:

$$\text{logits}_{\text{weak}} = \text{Softmax}(\text{FC}(\text{Avg}(f_e))) \quad (15)$$

After obtaining smoother non-target logits different from the original final logits, we apply a smoothing operation to the original labels $Labels$ to obtain smoothed labels $Labels_V$. Finally, we use the smoothed labels $Labels_V$ and the weak logits $\text{logits}_{\text{weak}}$ to compute the weak logits loss $\text{Loss}_{\text{weak}}$. The above process can be represented as:

$$Labels_V = \text{Smoothing}(Labels) \quad (16)$$

$$\text{Loss}_{\text{weak}} = -\mu \cdot \sum_i^C Labels_{Vi} \cdot \text{Log}(\text{logits}_{\text{weak}i}) \quad (17)$$

μ ($\mu < 1$) here is a hyperparameter utilized for implementing weak supervision. Log denotes the logarithmic operation.

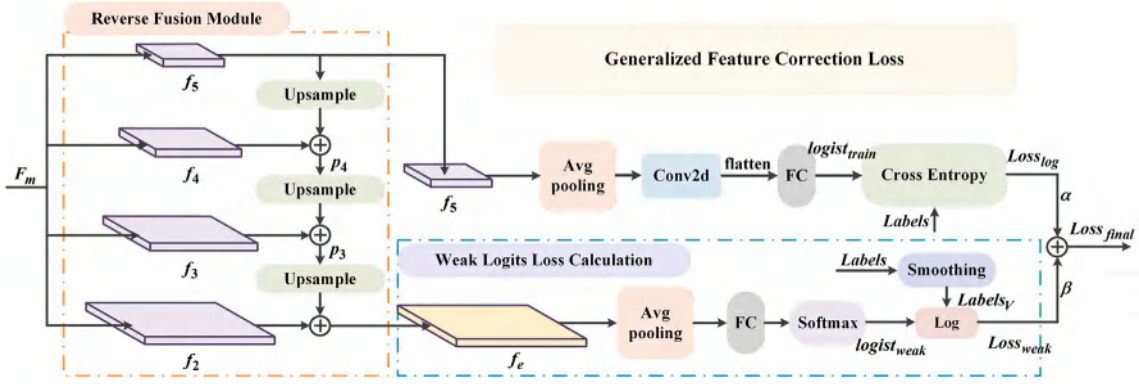


Fig. 6. The structure of generalized feature correction loss.

Algorithm 2: GFC Loss.

Input:
 F_m : the intermediate features of the model ;
 $Labels$: original labels ;

Output:
 $Loss_{final}$: final output loss ;

- 1 $f_5, f_4, f_3, f_2 \leftarrow F_m$; f_5, f_4, f_3, f_2 are constituents of F_m ;
- 2 $p_5 \leftarrow f_5$;
- 3 **for** $i = 5$ to 3 **do**
- 4 $k_{i-1} \leftarrow \text{Up}(p_i)$; k_{i-1} is an intermediate variable ;
- 5 $p_{i-1} \leftarrow k_{i-1} + f_{i-1}$;
- 6 **end**
- 7 $f_e \leftarrow p_2$; p_2 serves as the fused output feature of the Reverse Fusion Module ;
- 8 $logits_{train} \leftarrow \text{FC}(\text{Co}(\text{Avg}(f_5)))$: The original logits are computed ;
- 9 $Loss_{log} \leftarrow \text{Cross_entropy}(logits_{train}, Labels)$: The original cross-entropy loss is computed ;
- 10 $logits_{weak} \leftarrow \text{Softmax}(\text{FC}(\text{Avg}(f_e)))$: The weak logits of the model are obtained ;
- 11 $Labels_V \leftarrow \text{Smoothing}(Labels)$: The original labels are smoothed ;
- 12 $Loss_{weak} \leftarrow -\mu \cdot \sum_i Labels_{V_i} \cdot \text{Log}(logits_{weak_i})$: The weak logits loss is computed ;
- 13 $Loss_{final} \leftarrow \alpha Loss_{log} + \beta Loss_{weak}$: α and β are the smoothing hyperparameters of the Generalized Feature Correction Loss;
- 14 **return** $Loss_{final}$

Table 2

The data distribution of the VDC Dataset.

Class	Training set images	Validation set images
deformation	4436	1478
dislocation	2749	916
glass breakage	914	304
glass crack	933	311
missing	2683	894
scratch	4336	1445
scratches	4116	1371
tearing	2684	894
all	22,851	7613

Table 3

The data distribution of GC10-DET.

Class	Training set images	Validation set images
punching (Pu)	153	66
welding line (Wl)	191	82
crescent gap (Cg)	158	68
water spot (Ws)	202	87
oil spot (Os)	143	61
skil spot (Ss)	456	195
inclusion (In)	151	65
rolled pit (Rp)	22	9
crease (Cr)	37	16
waist folding (Wf)	105	45
all	1618	694

GFC Loss fuses fine-grained features obtained in the backbone network into a generalized weak representation of the features themselves. It further combines smoothed labels to compute weak logits loss, which modifies the parameters of the backbone network. This approach facilitates deeper learning of fine-grained level features extracted by the backbone, effectively enhancing the network's generalization capability.

4. Experiments and analysis

This section introduces the relevant details of the experiment, including the dataset composition, evaluation indices, experimental details, ablation results, visualization outcomes, and comparison findings.

4.1. Dataset

4.1.1. Vehicle damage classification dataset

The Vehicle Damage Classification Dataset (VDC Dataset) is employed in this study to evaluate the proposed network. The images in this dataset were captured from the daily insurance business of insurance companies. It should be noted that all images underwent strict anonymization after collection, with all personal identity information

(such as license plates and faces) and insurance company-related sensitive information removed. This dataset contains vehicle images of various colors, brands, and models, captured from different angles (front, rear, left, and right). It encompasses eight damage categories: deformation, dislocation, glass breakage, glass crack, missing, scratch, scratches, and tearing. This ensures that the VDC Dataset closely aligns with the application scenarios of vehicle damage assessment, and its use enables an objective evaluation of our model's effectiveness in the domain of vehicle damage.

In response to the demand for vehicle intelligent damage assessment, we divide the VDC Dataset into 22,851 training set images and 7613 validation set images. Table 2 shows the names of the 8 vehicle damage classes and the number of images for each class in the training and validation sets.

4.1.2. Metallic surface defect datasets

In this study, the publicly available metallic surface defect dataset GC10-DET [66] is utilized to evaluate the generalization capability of our model in industrial applications. GC10-DET is a surface defect dataset collected from real-world industrial scenarios. The GC10-DET comprises 10 distinct defect classes, as outlined in Table 3. In contrast

to the VDC Dataset, GC10-DET features a more limited dataset size and primarily consists of grayscale images, mirroring the common characteristics of defect datasets. Hence, the employment of GC10-DET enables an impartial evaluation of the suitability and generalizability of our model in surface defect applications.

We divide the defect images in GC10-DET into a 7:3 ratio, resulting in a final set of 1618 training images and 694 validation images. Table 3 shows the names of the 10 defect classes and the number of images for each class in the training and validation sets.

4.2. Evaluation metrics

To evaluate the effectiveness of our model in vehicle damage classification and its applicability in various industrial settings, we employ standard performance metrics including precision and accuracy for individual classes, overall Top-1 accuracy, and Top-5 accuracy. Precision for each class and accuracy for each class are defined as follows:

$$P_n = \frac{TP(n)}{TP(n) + FP(n)} \times 100\% \quad (18)$$

$$Acc_n = \frac{TP(n) + TN(n)}{TP(n) + FN(n) + FP(n) + TN(n)} \times 100\% \quad (19)$$

where $TP(n)$ (True Positive) refers to the case where class n is correctly predicted as class n . $FP(n)$ (False Positive) is when the model predicts other classes as class n . $FN(n)$ (False Negative) is a case where the model predicts class n to be other classes. P_n represents the precision of the n th class sample, reflecting the probability of the model predicting the results correctly among all samples predicted as class n by the model. Acc_n represents the accuracy of the n th class sample, reflecting the probability of the model predicting the results correctly among all samples of class n .

Both Top-1 accuracy and Top-5 accuracy are metrics used to assess the predictive performance of the model across all classes. Top-1 accuracy is determined based on whether the class assigned the highest confidence score by the model matches the true class of the sample. Similarly, Top-5 accuracy is calculated by verifying whether the classes corresponding to the top five confidence scores in the model's predictions align with the true class of the sample.

4.3. Experimental details

The study utilizes the NVIDIA 3090 professional accelerator card for both training and testing. The operating system used was Ubuntu 18.04.6 LTS, with training acceleration facilitated by CUDA 12.2. Python 3.8 was utilized as the programming language, in combination with the PyTorch neural network development framework.

We employ a single GPU training approach with a batch size of 32, utilizing the AdamW algorithm with a momentum of 0.9 and a weight decay of 0.03. The initial learning rate is set to 0.000035, with a warmup learning rate of 0.000001. The hyperparameter μ in the Generalized Feature Correction Loss is set to 0.05, while the parameters α and β are set to 0.5 and 1, respectively. Ablation experiments related to parameter settings are presented and analyzed in Section 4.4.

All input images are resized to 224×224 as the model input. An improved RandAugment algorithm is adopted for data augmentation, with the number of operations $N=7$, meaning that seven different image transformation operations are randomly selected from the augmentation operation pool in each iteration. The magnitude standard deviation $m_std=0.5$ controls the randomness of the augmentation magnitude. Additionally, CutMix is used to generate new training samples through a cut-and-paste approach, forcing the model to focus on local discriminative features of the images.

4.4. Parametric ablation experiment

For the hyperparameters α and β in the Generalized Feature Correction Loss, we conduct ablation experiments with different values

Table 4

The ablation of different hyperparameters in the GFC loss.

α	β	Top-1 acc (%)	Top-5 acc (%)
0.25	0.25	80.90	98.90
0.5	0.25	80.47	98.74
1	0.25	80.69	98.75
0.25	0.5	80.36	98.71
0.5	0.5	80.45	98.84
1	0.5	80.48	98.96
0.25	1	80.27	98.74
0.5	1	81.05	98.79
1	1	80.84	98.84

Table 5

The ablation of all classes on the VDC Dataset by FGDANet.

Baseline	FGDANet				Top-1 acc (%)	Top-5 acc (%)
	IR Layer	RL Block	GFS Module	GFC Loss		
✓					79.21	98.82
✓	✓				82.07	99.05
✓		✓			81.26	98.95
✓			✓		80.94	98.90
✓				✓	81.05	98.79
✓	✓	✓	✓	✓	83.53	99.03

as shown in Table 4. From the experimental results, it can be concluded that the GFC Loss has the best corrective effect on the model when α is set to 0.5 and β is set to 1.

4.5. Module ablation experiment

In this section, we conduct ablation studies to validate the contribution of each module in FGDANet for vehicle damage classification. As shown in Table 5, incorporating each of the four modules into the baseline model leads to a consistent improvement in Top-1 accuracy. Specifically, adding the IR Layer increases the accuracy by 2.86%, while integrating the RL Block yields a 2.05% gain. Furthermore, the inclusion of the GFS Module and GFC Loss results in improvements of 1.73% and 1.84%, respectively. When all modules are integrated, the complete FGDANet achieves a 4.32% increase in Top-1 accuracy over the baseline. These results demonstrate that each module, both individually and in combination, effectively enhances the model's performance in vehicle damage classification.

The precision of the FGDANet model for each damage class during ablation is detailed in Table 6. The precision of a single class represents the proportion of samples actually belonging to that class among the samples predicted as that class by the model. Table 7 provides the accuracy of the FGDANet model for each damage class during ablation. Single-class accuracy represents the proportion of correctly predicted samples of that class among the actual samples of that class. By combining Tables 6 and 7, the performance of each module can be more intuitively evaluated. It is evident from the tables that the four modules place different emphases on the feature focus areas for individual classes, leading to different improvements in the model's classification ability for different classes with the addition of different modules. Overall, the addition of each module can enhance the model's classification performance for different vehicle damage classes.

4.6. Visualization experiment

Through ablation experiments, we quantitatively analyzed the effectiveness of each module in our proposed model using metrics such as precision and accuracy. To visually validate the effectiveness of our model in feature extraction for vehicle damage classification, we randomly selected 50 images per class (8 classes in total, 400 images) from the validation set. Deep features of these images were extracted using

Table 6
The ablation of single-class precision on the VDC Dataset by FGDA Net.

Baseline	FGDA Net				Precision (%)							
	IR Layer	RL Block	GFS Module	GFC Loss	deformation	dislocation	glass breakage	glass crack	missing	scratch	scratches	tearing
✓					81.08	80.06	85.93	75.38	68.87	88.34	73.57	79.10
✓	✓				80.41	86.63	91.51	86.03	75.10	89.21	77.92	77.45
✓		✓			83.36	86.21	78.79	85.00	75.79	88.27	75.43	75.09
✓			✓		82.59	83.88	87.55	82.38	73.03	89.58	75.81	73.80
✓				✓	81.74	84.61	87.77	82.50	77.54	86.58	74.28	77.50
✓	✓	✓	✓	✓	85.05	86.15	84.28	86.64	76.93	90.49	79.05	78.34

Table 7
The ablation of single-class accuracy on the VDC Dataset by FGDA Net.

Baseline	FGDA Net				Accuracy (%)							
	IR Layer	RL Block	GFS Module	GFC Loss	deformation	dislocation	glass breakage	glass crack	missing	scratch	scratches	tearing
✓					82.61	84.61	76.32	63.02	79.42	92.80	72.28	63.09
✓	✓				88.29	84.17	81.58	63.34	80.65	94.95	73.09	70.69
✓		✓			85.05	81.22	85.53	60.13	80.54	94.26	76.37	68.12
✓			✓		85.72	83.52	78.62	69.13	79.98	93.36	73.81	67.11
✓				✓	86.00	82.21	80.26	63.67	77.63	94.67	74.98	68.57
✓	✓	✓	✓	✓	85.05	86.24	88.16	73.00	77.96	95.43	77.90	71.59

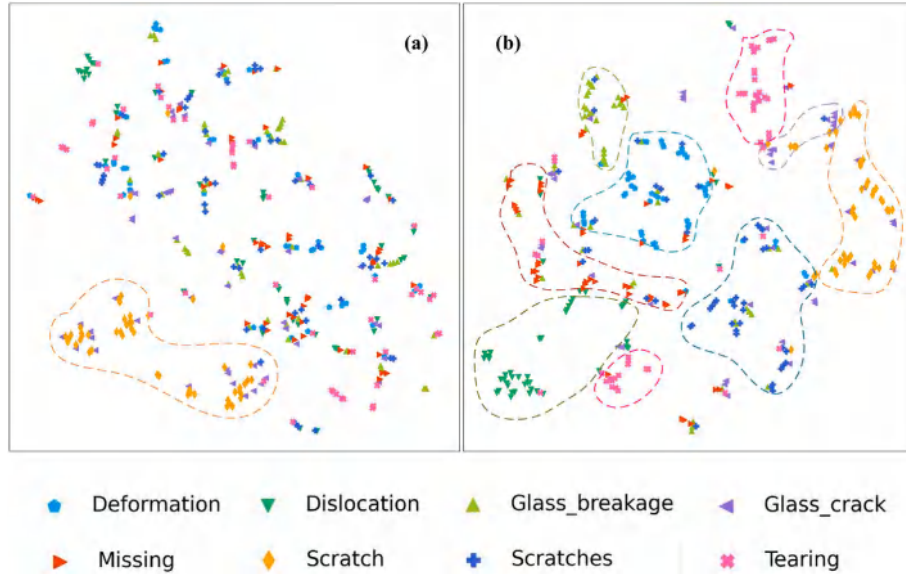


Fig. 7. The t-SNE feature dimensionality reduction results of the baseline and FGDA Net: (a) features extracted by the baseline from the images, (b) features extracted by FGDA Net from the images.

both the baseline model and our proposed model, and then dimensionality reduction was performed using t-SNE [67] to project the features onto a two-dimensional plane for visualization, as shown in Fig. 7. If the features extracted by our model exhibited smaller intra-class distances and clearer inter-class boundaries compared to the baseline, it would demonstrate the superior discriminability of our model. Analysis of Fig. 7(a) reveals that the baseline model extracted various damage features that exhibited significant confusion, with features such as deformation, dislocation, glass breakage, and missing showing dispersed distributions, intermingling with each other, and lacking clear clustering. In contrast, FGDA Net, leveraging its four feature perception modules, effectively captured the fine-grained features of the damaged areas in the images, thereby reducing feature confusion. As illustrated in Fig. 7(b), the damage features extracted by FGDA Net exhibited better intra-class cohesion and inter-class segregation compared to the baseline model. Consequently, it can be concluded from the feature level that our model pays fine-grained attention to the key areas of various

vehicle damage types, and it exhibits a significantly improved ability to learn the features of vehicle damage compared to the baseline model.

Simultaneously, representative images from each class of the validation set are selected and input into both the baseline model and our model. The Grad-CAM algorithm [68] is utilized to visualize the locations of the model's attention features. Fig. 8 displays the class activation maps of the baseline model and FGDA Net when predicting images of eight types of vehicle damage. It is evident from Fig. 8 that, in comparison to the baseline model, FGDA Net can better focus on the discriminative regions belonging to the damage class, effectively learning the fine-grained features of vehicle damage areas.

4.7. Comparison experiment

Table 8 provides a detailed comparison of the classification precision and accuracy of the baseline network and FGDA Net for the 8 types of vehicle damage on the VDC dataset. The results demonstrate that FGDA Net significantly outperforms the baseline model in

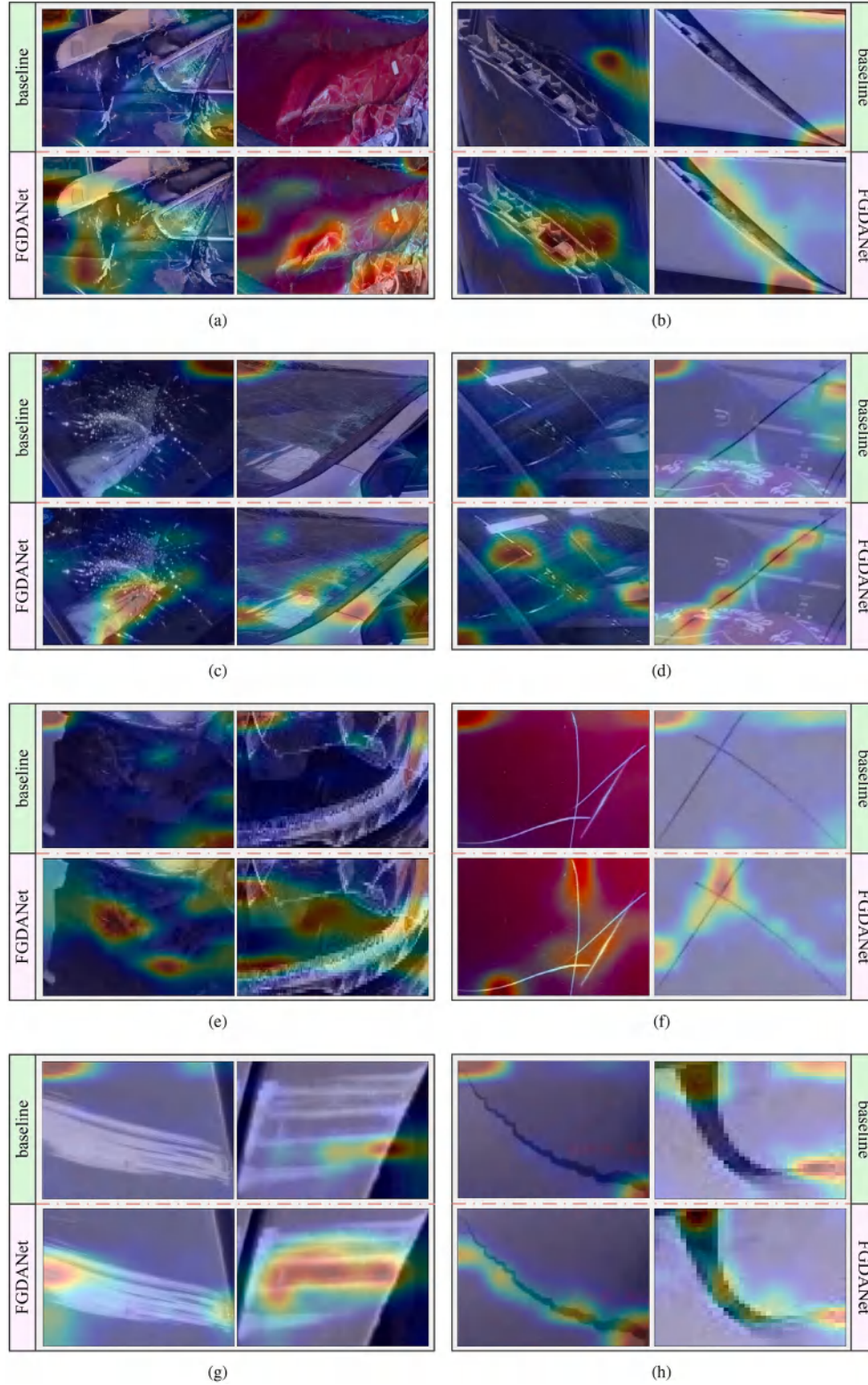


Fig. 8. Comparison of Grad-CAM visualization results for eight types of damage on baseline and FGDA Net: two different representative data images are shown for each class, (a) for deformation, (b) for dislocation, (c) for glass breakage, (d) for glass crack, (e) for missing, (f) for scratch, (g) for scratches, and (h) for tearing.

classifying vehicle damage. Specifically, for the “glass crack” class, FGDA Net exhibits an improvement of 8.62% in classification precision and 11.26% in accuracy compared to the baseline model. When evaluating based on individual class precision, FGDA Net demonstrates better classification performance for 6 types of damage compared to the

baseline model. Similarly, when evaluating based on individual class accuracy, FGDA Net exhibits better classification performance for 7 types of damage compared to the baseline model. The comparative experiment between the baseline model and our proposed model validates the overall improvement of FGDA Net in the classification of vehicle damage.

Table 8

The comparison of class precision and accuracy between the baseline and FGDANet.

Class	Baseline		FGDANet	
	Precision (%)	Accuracy (%)	Precision (%)	Accuracy (%)
deformation	81.08	82.61	85.05	85.05
dislocation	80.06	84.61	86.15	86.24
glass breakage	85.93	76.32	84.28	88.16
glass crack	75.38	63.02	86.64	73.00
missing	68.87	79.42	76.93	77.96
scratch	88.34	92.80	90.49	95.43
scratches	73.57	72.28	79.05	77.90
tearing	79.10	63.09	78.34	71.59

To further demonstrate the superiority of our model in vehicle damage classification, we conduct a comparative analysis of FGDANet with advanced image classification algorithms, using the VDC Dataset. The results are presented in Table 9. It is evident from the table that several networks, renowned for their robust classification capabilities in natural scene objects, encounter challenges in discerning the distinctive features of damaged areas within the VDC Dataset. The difficulty of distinguishing between classes and the complexity of morphology within classes hinder these networks from achieving fine-grained learning of the key region features, consequently leading to their subpar performance in vehicle damage classification.

The proposed FGDANet exhibits a notable advantage in the classification of vehicle damage compared to other state-of-the-art image classification networks. As shown in Table 9, our method achieves a Top-1 accuracy of 83.53% on the VDC Dataset. This represents an improvement of 3.63% and 4.23% over the state-of-the-art CNN-based models FasterNet [13] and StartNet [75], respectively; an improvement of 8.78% and 9.75% over the advanced Transformer-based models MALA [78] and RALVT [79], respectively; and an improvement of 2.59% and 2.96% over the advanced Mamba-based models VMamba [80] and VSSD [81], respectively. The comparative experiments with other classification networks on the VDC Dataset effectively confirm the superior performance and effectiveness of FGDANet in vehicle damage classification.

Table 9 also presents a comparison of computational efficiency among different methods on the VDC Dataset. It can be observed that,

Table 9

The comparison of classification results between FGDANet and other classification models on the VDC Dataset.

Method	Type	Top-1 acc (%)	Top-5 acc (%)	FLOPs(G)	Params(M)
AlexNet [7]	CNN	65.36	97.22	0.3	15
GoogleNet [69]	CNN	67.86	97.63	1.6	7
MobileNet [11]	CNN	65.62	97.28	0.3	4
DenseNet [70]	CNN	79.85	99.11	2.9	8
ShuffleNet [71]	CNN	78.21	99.07	0.2	2
EfficientNet [72]	CNN	80.70	99.00	0.4	5
RegNet [73]	CNN	80.34	99.01	0.2	2
EfficientNetv2 [12]	CNN	81.61	99.23	2.9	21
RepLKNNet [74]	CNN	80.94	98.95	15.3	79
FasterNet [13]	CNN	79.90	99.03	4.5	31
StartNet [75]	CNN	79.30	98.79	0.4	3
Vision Transformer [8]	Transformer	71.93	98.57	4.4	88
Swin Transformer [35]	Transformer	80.03	99.09	4.4	28
Metaformer [76]	Transformer	75.77	98.77	3.8	24
RMT [77]	Transformer	75.58	98.34	4.5	26
MALA [78]	Transformer	74.75	98.15	4.6	27
RAVLT [79]	Transformer	73.78	98.03	4.6	26
VMamba [80]	Mamba	80.94	99.08	4.8	31
VSSD [81]	Mamba	80.57	99.08	4.5	24
Baseline [10]	CNN	79.21	98.82	4.1	26
Ours	CNN	83.53	99.03	4.8	35

Table 10

The comparison of classification results between FGDANet and other classification models on the GC10-DET.

Method	Type	Top-1 acc (%)	Top-5 acc (%)
AlexNet [7]	CNN	47.65	91.61
GoogleNet [69]	CNN	52.35	93.92
ResNet [10]	CNN	54.68	91.16
MobileNet [11]	CNN	47.98	88.33
DenseNet [70]	CNN	50.14	84.01
ShuffleNet [71]	CNN	52.45	85.74
EfficientNet [72]	CNN	51.15	91.79
RegNet [73]	CNN	50.29	90.92
EfficientNetv2 [12]	CNN	52.31	89.34
RepLKNNet [74]	CNN	45.68	92.22
FasterNet [13]	CNN	51.59	90.35
StartNet [75]	CNN	51.61	91.64
Vision Transformer [8]	Transformer	48.70	87.90
Swin Transformer [35]	Transformer	54.37	89.60
Metaformer [76]	Transformer	38.90	83.14
RMT [77]	Transformer	53.75	90.92
MALA [78]	Transformer	50.58	91.07
RAVLT [79]	Transformer	51.87	88.90
VMamba [80]	Mamba	45.82	91.64
VSSD [81]	Mamba	53.17	92.94
Ours	CNN	58.65	94.38

compared to some baseline models, our method incurs a slight increase in parameter count and computational cost. However, this represents a reasonable trade-off considering the improvement in classification accuracy achieved by the model. These modest computational costs are acceptable for the task of intelligent vehicle damage assessment.

4.8. Generalization experiment

In order to provide further evidence of the learning capability of the proposed network for defect target features and to validate its generalization, this study compares the defect classification accuracy of FGDANet with other state-of-the-art image classification algorithms using the GC10-DET, as presented in Table 10. The results clearly demonstrate that FGDANet outperforms other classification algorithms in accurately classifying steel plate surface defects.

The proposed FGDANet, through improvements in the feature extraction capability of the backbone network, effectively learns features

from strongly discriminative regions of defect targets, achieving fine-grained classification of defects. On GC10-DET, FGDANet achieves a Top-1 accuracy of 58.65%. In contrast, existing classification networks based on the CNN framework struggle to effectively classify defect targets due to their limited fine-grained feature extraction capabilities. Experimental results demonstrate that among CNN-based models, ResNet achieves the best Top-1 accuracy of 54.68%, which is 3.97% lower than that of FGDANet. Classification networks based on the Transformer framework, on the other hand, are constrained by the limited size of the dataset, making it difficult for them to meet the requirements of industrial applications for defect classification. Experimental results show that among Transformer-based models, Swin Transformer achieves the best Top-1 accuracy of 54.37%, which is 4.28% lower than that of FGDANet. Classification networks based on the Mamba framework struggle to effectively model subtle local differences, also hindering accurate classification. Experimental results indicate that among Mamba-based models, VSSD achieves the best Top-1 accuracy of 53.17%, which is 5.48% lower than that of FGDANet. The comparative experiments in Table 10 fully demonstrate the generalization capability of FGDANet in defect classification applications.

5. Conclusion

This paper addresses the challenges of high inter-class similarity and large intra-class variance in vehicle damage classification by constructing a solution centered on fine-grained feature perception. At the methodological level, four modular designs are introduced to achieve selective enhancement and multi-level fusion of damaged region features: the alternating introduction of the IR Layer and RL Block mitigates information loss during downsampling, enabling the model to focus on subtle structures in critical regions; the GFS Module incorporates shallow global self-similar information into the discriminative process, enhancing inter-class discrimination; and the GFC Loss imposes regularization constraints on the backbone network through reverse feature fusion and label smoothing strategies, improving feature generalization. At the experimental level, systematic ablation studies and comparative analyses are conducted on the VDC dataset to validate the effectiveness of each module, and the cross-domain generalization capability of the model is examined on the GC10-DET dataset. Experimental results demonstrate that the proposed method outperforms current mainstream classification networks on damage defect classification tasks, with notable improvements in feature clustering and attention localization accuracy. The aforementioned work indicates that network architecture design and loss optimization strategies oriented toward fine-grained feature perception represent an effective pathway for improving industrial defect classification performance.

Although FGDANet has achieved significant progress in damage defect classification tasks, the current method focuses solely on the classification task of damage images and fails to simultaneously localize damage positions and components. Future work will explore multi-task learning frameworks, enabling the model to output both damage region masks and corresponding component masks during classification, thereby forming an integrated “classification-detection-segmentation” damage assessment pipeline. Furthermore, defect samples in industrial scenarios are often difficult to acquire and exhibit continuously evolving distributions. Subsequent research will investigate domain generalization and few-shot learning methods to enhance the model’s rapid adaptability to unseen defect categories and novel damage morphologies, further strengthening its practical value in real-world claims assessment scenarios.

CRedit authorship contribution statement

Yongjie Zhai: Writing – review & editing, Supervision, Project administration, Conceptualization. **Zhenqi Zhang:** Writing – original draft, Supervision, Software, Methodology, Data curation, Writing – review & editing. **Xunqi Zhou:** Visualization, Validation,

Investigation, Formal analysis. **Zixuan Wang:** Investigation, Data curation, Conceptualization. **Qianming Wang:** Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work is supported in part by the [Fundamental Research Funds for the Central Universities \(2023JC006\)](#), the [National Natural Science Foundation of China \(NSFC\)](#) under grant number 62373151, the [Hebei Province Natural Science Foundation Surface Project](#) under grant number F2023502010, and the [Fundamental Research Funds for the Central Universities](#) under grant number 2024MS136.

Data availability

The data that has been used is confidential.

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